Problem set 4
Probabilistic classifiers

Exercise 1

1. For Example 2 in "Probabilistic classifiers" video, compute the maximum likelihood estimation for the parameter β when the observed data contains h heads and t tails.

   Hint*: find β that maximizes the log likelihood function $h \log \beta + t \log(1 - \beta)$

2. Interpret the resulting estimate.

Excercise 2

Feature x can take one of the three values: cat, dog, rabbit. Let $n(\text{cat}, c) = 9$, $n(\text{dog}, c) = 0$, and $n(\text{rabbit}, c) = 1$, where $n(a, c)$ denotes the number of training objects in class c with $x = a$.

1. Estimate the probabilities $P(x = \text{cat} \mid c)$, $P(x = \text{dog} \mid c)$, $P(x = \text{rabbit} \mid c)$.

2. Smooth the obtained probabilities using Laplace smoothing.

Excercise 3

Imagine that you are given the following set of training examples. Each feature can take on one of three nominal values: a, b, or c.

F_1	F_2	F_3	Class Label
a	c	b	+
c	a	c	+
a	a	c	-
b	c	a	-
c	c	b	-

How would a Naive Bayes system classify the following test example? $F_1 = a, F_2 = c, F_3 = b$

Excercise 4

1% of women at age forty who participate in routine screening have breast cancer. 80% of women with breast cancer will get positive mammographies. 9.6% of women without breast cancer will also get positive mammographies. A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has breast cancer?

Excercise 5

There is a 50% chance there is both life and water on Mars, a 25% chance there is life but no water, and a 25% chance there is no life and no water. What is the probability that there is life on Mars given that there is water?

Excercise 6

In a medical study, 100 patients all fell into one of three classes: Pneumonia, Flu, or Healthy. The following database indicates how many patients in each class had fever and headache.

Pneumonia		
Fever	Headache	Count
T	T	5
T	F	0
F	T	4
F	F	1

What values would a naive Bayes classifier assign to the three possible diagnoses? Show your work. (For this question, the three values need not sum to 1. Recall that the naive Bayes classifier drops the denominator because it is the same for all three classes.)

Excercise 7

What is an important drawback of maximum likelihood estimation?

Excercise 8

We want to build a naive bayes sentiment classifier using add laplace smoothing, as described in the lecture. Here is our training corpus:

- the movie has no plot
- honestly pretty boring
+ pretty interesting movie

Compute the prior for the two classes + and -, and the likelihoods for each word given the class (leave in the form of fractions).

Excercise 9

In machine learning classification, the *discriminative approach* models the conditional probability of the class label given the observed features, i.e.,

$$P(y \mid x),$$

rather than modeling the joint distribution $P(x, y)$ or the class-conditional likelihood $P(x \mid y)$. State whether this statement is **True** or **False**.

Excercise 10

Applying Naïve Bayes to data with numerical attributes and using the Laplace correction Given the training data in the table below (Tennis data with some numerical attributes), predict the class of the following new example using Naïve Bayes classification: outlook=overcast, temperature=60, humidity=62, windy=false.

outlook	**temperature**	**humidity**	**windy**	**play**
sunny	85	85	false	no
sunny	80	90	true	no
overcast	83	86	false	yes
rainy	70	96	false	yes
rainy	68	80	false	yes
rainy	65	70	true	no
overcast	64	65	true	yes
sunny	72	95	false	no
sunny	69	70	false	yes
rainy	75	80	false	yes
sunny	75	70	true	yes
overcast	72	90	true	yes
overcast	81	75	false	yes
rainy	71	91	true	no

Table 1: Play Tennis Dataset